IMO Training
Combinatorics

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Topic 1 Counting

1.1 The four arithmetic operations

We frequently use the four basic arithmetic operations in counting. Can you give examples?

- Addition rule and multiplication rule — the two very basic operations
- Subtraction — counting the opposite
- Division — to correct for overcounting

Another reason for subtraction is to correct for overcounting, summarised by the **inclusion-exclusion principle**.

1.2 Permutations of *n* objects

How many **permutations** of *n* objects are there if

- the objects are distinct?
- some of the objects are the same?
- we only need to permute *some* (rather than *all*) of the objects?
- the permutation is *circular*?

1.3 The '*n* choose *r*' problem

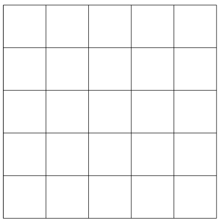
How many ways are there to choose *r* objects out of *n* (distinct) objects?

	Order matters	Order does not matter
Repetition allowed	(a)	(b)
Repetition not allowed	(c)	(d)

1.4 Miscellaneous examples

1. (PCIMC 2003) How many 4-digit positive integers have the product of their digits even?

2. The figure shows a 5×5 square grid.



   (a) How many squares can be found?

   (b) How many rectangles (including squares) can be found?

   (c) What is the sum of the areas of the rectangles in (b)?

   (d) In how many ways can one travel from the lower left hand corner to the upper right hand corner, if in each step one can only go upward or rightward?

3. There are some red, yellow, green and blue marbles. There are 20 marbles of each colour.

   (a) If we need to choose two marbles of different colours, how many choices are there?

   (b) If we need to choose ten marbles, how many different combinations of colours are there?

   (c) What will be the answer to (b) if we require at least two red marbles?

4. How many ways are there to divide a class of 40 students into eight groups of 5?

5. In how many ways can we choose 13 cards from a pack of 52 so that there is at least one void?

6. In the Mark Six lottery in Hong Kong, what is the probability of winning the various prizes?

7. When 5 cards are drawn from a pack of 52, what is the probability of getting the various combinations?

Exercise 1

1. How many ways are there to seat 8 boys and 4 girls at a round table if no two girls may sit next to each other?

2. Six men and six women are to be paired up. Two men and two women insist pairing up with someone of the same gender. Another two men and two women insist pairing up with someone of the opposite gender. How many ways of pairing are there?

3. (IMO HK Prelim 1999) A solid pyramid *VABCD*, with a quadrilateral base *ABCD*, is to be coloured on each of the five faces such that no two faces with a common edge will have the same colour. If five different colours are available, what is the number of ways to colour the pyramid?

4. (AIME 2006) Let $(a_1, a_2, a_3, \ldots, a_{12})$ be a permutation of (1, 2, 3, …, 12) for which

$$a_1 > a_2 > a_3 > a_4 > a_5 > a_6 \text{ and } a_6 < a_7 < a_8 < a_9 < a_{10} < a_{11} < a_{12}.$$

An example of such a permutation is (6, 5, 4, 3, 2, 1, 7, 8, 9, 10, 11, 12). Find the number of such permutations.

5. (AIME 2008, modified) There are two distinguishable flagpoles, and there are 19 flags, of which 10 are identical blue flags, and 9 are identical green flags. In how many different ways can we put all the flags onto the flagpoles in which each flagpole has at least one flag and no two green flags on either pole are adjacent?

6. (PCIMC 2008) Find the number of 25-digit positive integers satisfying the following two conditions:

   - The digits of the integer consist of exactly 20 copies of ‘1’ and 5 copies of ‘2’.
   - If we remove the last *k* digits from the integer (where *k* is any positive integer less than 25), the resulting integer has at least as many ‘1’s as ‘2’s.

7. (CMO 2001) On a circle with circumference 24 there are 24 equally spaced points, marked 1 to 24. In how many ways can we choose 8 out of these 24 points so that the length of the minor arc joining any two of the chosen points is not equal to 3 or 8?

8. (CGMO 2008) Given a 2008×2008 chessboard, each cell of which has a different colour. Now one of the 4 letters C, G, M, O is to be filled into each cell. If every 2×2 grid on the chessboard contains all 4 letters C, G, M, O, we say that the chessboard is 'harmonic'. How many different 'harmonic' chessboards are there?

Topic 2 Existence and Optimisation

2.1 The pigeonhole principle

The basic form of the **pigeonhole principle** says

- If there are $n+1$ pigeons and n holes, then ____.

What are some alternative forms?

2.2 Existence and sharpness of bound

Using the pigeonhole principle, we can show that any 51 numbers chosen from the set {1, 2, …, 100} must satisfy certain *existence* properties. Some possible properties are:

- Two of the chosen numbers have sum 101.
- Two of the chosen numbers are relatively prime.
- One of the chosen numbers divides another chosen number.

In each case, determine whether the bound 51 is *sharp*.

2.3 Structure of an optimisation problem

An *optimisation* problem generally refers to one which asks for the maximum/minimum value of a quantity subject to certain conditions. In general, we need to do three things in answering such a problem:

- Find the answer
- Show that the answer you claimed satisfies the required conditions
- Show the the answer you claimed is indeed the maximum/minimum

(IMO 1977) In a finite sequence of real numbers the sum of any seven successive terms is negative, and the sum of any eleven successive terms is positive. Determine the maximum number of terms in the sequence.

(APMO 1992) Find a sequence of maximal length of non-zero integers in which the sum of any seven consecutive terms is positive and that of any eleven consecutive terms is negative.

2.4 Miscellaneous examples

1. In a square of side length 4, is it possible to choose 28 points so that no circle with centre inside the square and radius 1 covers 4 or more points?

2. Given nine straight lines, each of which divides a square into two quadrilaterals of areas $2:3$. Show that three of these lines meet at a point.

3. (IMO 1972) Prove that from a set of ten distinct two-digit numbers (in the decimal system), it is possible to select two disjoint subsets whose members have the same sum.

4. At least how many integers must be chosen to ensure that there must be two chosen integers who sum or difference is divisible by 1000?

Exercise 2

1. In a bag there are 1000 balls, numbered 1 to 1000. What is the minimum number of balls that must be drawn from the bag to ensure that you must be able to get two balls whose sum of numbers has unit digit 7?

2. Nine different real numbers are chosen. Show that we can find x and y among them such that $0 < \frac{x-y}{1+xy} < \sqrt{2}-1$.

3. (JMO 1991) In every 16-digit number, show that there is a string of one or more consecutive digits such that the product of these digits is a perfect square.

4. (IMO 2002 HK Selection Test) For an irrational number x, let x' be the integer nearest to x. Define $\langle x \rangle = |x - x'|$. Show that for every irrational number y, the minimum of the numbers $\langle y \rangle$, $\langle 2y \rangle$, …, $\langle 2001y \rangle$ is less than $\frac{1}{2001}$.

5. In a school, there are 601 students, no two of whom have the same height. They are queued up in a row in random order. Show that the teacher can always find at least one of the following:
   - 21 students in the queue whose heights are in ascending order; or
   - 31 students in the queue whose heights are in descending order.

6. (PCIMC 2019) There is a $2 \times n$ table. In each cell a different positive integer is written. If we add up the two numbers in each column, the sums obtained are 2019, 2020, …, $2018+n$ respectively. Find the greatest possible value of n.

7. (AIME 2010) Let $m \geq 3$ be an integer and let $S = \{3, 4, 5, \ldots, m\}$. Find the smallest value of m such that for every partition of S into two subsets, at least one of the subsets contains integers a, b and c (not necessarily distinct) such that $ab = c$.

8. (CHKMO 1999, rephrased) In a class each student studies the same n subjects. One day there is a test in each subject. It is known that for each subject exactly three students fail the test, and for any two distinct subjects exactly one student fail both tests. Determine the smallest possible value of n for which the above conditions imply that exactly one student fails all n tests.

9. (IMO 2012 shortlist) Let $n \geq 1$ be an integer. What is the maximum number of disjoint pairs of elements of the set $\{1, 2, \ldots, n\}$ such that the sums of the different pairs are different integers not exceeding n?

Topic 3 Recurrence Relations

3.1 Examples for motivation

1. (IMO HK Prelim 1991) A flight of stairs consists of 10 steps. A boy can climb one or two steps each time. In how many different ways can the boy climb from the bottom to the top of the stairs?

2. (PCIMC 2006) In a city, all telephone numbers have 8 digits and all telephones have a keyboard in the form as shown. Furthermore, consecutive digits in a telephone number must be adjacent digits on the keyboard (e.g. 85256321). How many different possible telephone numbers are there?

7	8	9
4	5	6
1	2	3

3. (IMO HK Prelim 2001) Find the number of 10-digit positive integers such that

   (a) each digit is either 1 or 2; and

   (b) there exist two consecutive 1's.

4. (AIME 2008) Consider sequences that consist entirely of *A*'s and *B*'s and that have the property that every run of consecutive *A*'s has even length, and every run of consecutive *B*'s has odd length. Examples of such sequences are *AA*, *B*, and *AABAA*, while *BBAB* is not such a sequence. How many such sequences have length 14?

3.2 Solving recurrence relations (I) — easy cases

Solving a recurrence relation means we want to find the _____________________ of the sequence.

Some recurrence relations are easy to solve.

- Write down an arithmetic sequence and a recurrence relation that it satisfies.
- Write down a geometric sequence and a recurrence relation that it satisfies.
- Can you solve the above recurrence relations? What is the role of the **initial conditions**?
- How do the initial conditions for the Fibonacci sequence differ from those for arithmetic and geometric sequences?
- Can you find a second order recurrence relation for a geometric sequence?
- Can you solve the recurrence relation $a_{n+1} = 3a_n - 4$ with $a_1 = 5$?

3.3 Solving recurrence relations (II) — linear homogeneous with constant coefficients

Linear homogeneous recurrence relations with constant coefficients can be solved by the **method of characteristic equations**.

- $a_{n+2} = 4a_{n+1} - 3a_n$ with $a_1 = 1$ and $a_2 = 2$
- $a_{n+2} = 4a_{n+1} - 4a_n$ with $a_1 = 1$ and $a_2 = 3$

3.4 Solving recurrence relations (III) — when a non-homogeneous term is added

In this case, the general solution is the sum of the general solution to the homogeneous part and a particular solution.

- $a_{n+2} = 4a_{n+1} - 4a_n + 2n$ with $a_1 = 1$ and $a_2 = 3$
- $a_{n+2} = 4a_{n+1} - 4a_n + 3^n$ with $a_1 = 1$ and $a_2 = 3$
- $a_{n+2} = 4a_{n+1} - 3a_n + 3^n$ with $a_1 = 1$ and $a_2 = 3$

Exercise 3

1. Find the unit digit and the tenth digit (i.e. the two digits immediately next to the decimal point) of the number $(3+\sqrt{5})^{2345}$.

2. (IMO 2019 HK Team Selection Test) Determine all sequences p_1, p_2, p_3, … of prime numbers for which there exists an integer k such that the recurrence relation

$$p_{n+2} = p_{n+1} + p_n + k$$

holds for all positive integers n.

3. Let a_n be the number of positive integers having digits 1, 3, 4 only and sum of digits equal to n.

   (a) Find a_{12}.

   (b) Show that a_{2n} is a perfect square for all natural numbers n.

4. (IMO 1979) Let A and E be opposite vertices of a regular octagon. A frog starts jumping at vertex A. From any vertex of the octagon except E, it may jump to either of the two adjacent vertices. When it reaches vertex E, the frog stops and stays there. Let a_n be the number of distinct paths of exactly n jumps ending at E. Prove that $a_{2n-1} = 0$ and

$$a_{2n} = \frac{1}{\sqrt{2}}\left[\left(2+\sqrt{2}\right)^{n-1} - \left(2-\sqrt{2}\right)^{n-1}\right] \text{ for } n = 1, 2, 3, \ldots$$

5. (CGMO 2008) Let the sequence x_1, x_2, …, x_n, … of positive real numbers satisfy $(8x_2 - 7x_1)x_1^7 = 8$ and

$$x_{k+1}x_{k-1} - x_k^2 = \frac{x_{k-1}^8 - x_k^8}{(x_k x_{k-1})^7}$$

for $k \geq 2$. Find positive real number a such that, when $x_1 > a$, we have $x_1 > x_2 > \cdots > x_n > \cdots$, while when $0 < x_1 < a$, the sequence is not monotonic.

Topic 4 Invariants and Monovariants

4.1 Invariants

Many problems in combinatorics ask whether it is possible to reach a certain state given the initial state and the allowed operations. It is usually easier to argue for a *yes* since one only needs to provide the actual sequence of operations, whereas if the answer is *no* one must argue that no matter how one operates the desired state cannot be reached.

In the latter case it is often useful to look for an *invariant* to assist our proof. As its name implies, an invariant is a quantity which does not change during the operation. If such quantity has different values in the initial state and the desired state, then we know that it is not possible to reach the desired state.

The following is a classic example.

There are n lamps, initially all off. Each lamp is equipped with a switch. When pressed, a lamp off will be turned on and a lamp on will be turned off. One is allowed to press exactly 100 switches each time.

(a) When $n = 1234$, is it possible to have all the lamps on at the same time?

(b) When $n = 1235$, is it possible to have all the lamps on at the same time?

4.2 Monovariants and the extremal principle

Like an invariant, a monovariant is a quantity related to a combinatorics problem which will assist us in proving that the desired state cannot be reached. Unlike an invariant, however, the value of a monovariant may change after the allowed operations. Yet it always changes in the same direction (hence the prefix *mono*-), e.g. increasing.

Such a quantity can help us in various ways. For example, if we can identify such a quantity which is increasing after an operation and that the value of this quantity in the desired state is smaller than that in the initial state, then we know that it is not possible to reach the desired state. On the other hand, such a quantity can also be used to argue for a *yes* in terms of reaching a desired state, by arguing that the quantity is not minimised/maximised when the desired state has not been reached.

The following example illustrates the latter use of a monovariant. When used in such way, sometimes we say that the **extremal principle** has been employed.

In the IMO training programme, each student has exactly three friends (friendship is mutual). Prove that it is possible to divide the students into two levels so that each student finds at most one friend in the same level.

4.3 Miscellaneous examples

1. A positive integer starts with 9763… and each digit from the fifth is equal to the sum of the previous four digits modulo 10. (Thus the integer reads 97635154…) Does the integer contain 1234 as consecutive digits somewhere (provided it has sufficiently many digits)?

2. On an island there are 13 red birds, 15 yellow birds and 17 green birds. When two birds of different colours meet, both of them change into the third colour. Is it possible for all birds to change into the same colour?

3. On a board the numbers 4, 6, 8 are written. Each time one is allowed to erase two numbers a and b and replace them by $0.6a-0.8b$ and $0.8a+0.6b$. Is it possible for the numbers 5, 6, 7 to appear on the board at the same time?

4. (Putnam 2008 modified) On a board finitely many positive integers are written. Each time one is allowed to erase two integers provided that none of them is a multiple of the other, and then replace them by their L.C.M. and H.C.F. respectively. Is it possible for this process to continue forever?

5. On a board the numbers 1, 2, 3, 4, 5 are written. Each time one is allowed to erase two numbers a and b and replace them by $a+b$ and ab. Is it possible for the numbers 12, 123, 1234, 12345 and 123456 to appear on the board at the same time?

6. In each cell of an $m \times n$ table a real number is written. Each time one is allowed to multiply all numbers in a row or all numbers in a column by –1. Show that after a finite number of such operations one can make the sum of the entries in each row as well as in each column nonnegative.

Exercise 4

1. On a board the numbers 4, 6, 8 are written. Each time one is allowed to erase one of the three numbers, and replace it by one less than the sum of the other two numbers. (For example, one may erase the number 4 and replace it by $6+8-1=13$.) Is it possible for the numbers 10001, 20001 and 30001 to appear on the board at the same time?

2. There are 1000 cards, numbered 1 to 1000, placed in a row in order. In each step one swaps two adjacent cards. Is it possible that the cards are in their initial order after exactly 1001 steps?

3. On the board there is a quadratic polynomial $f(x)$. Each time one is allowed to exchange the constant term with the coefficient of x^2, or replace $f(x)$ by $f(x+t)$ for some real number *t*. If the initial polynomial is x^2-x-2, can we get x^2-x-1 in finitely many steps?

4. Some (finitely many) stones are placed on the real number line, each at a lattice point. At each step, one is allowed to choose a point with at least two stones, and then move one stone from that point to the left by 1 unit and another from that point to the right by 1 unit. Is it possible to return all the stones to the initial position after a finite (positive) number of steps?

5. A positive integer starts with 101010… and each digit from the seventh is equal to the sum of the previous six digits modulo 10. (Thus the integer reads 101010350985…) Does the integer contain 010101 as consecutive digits somewhere (provided it has sufficiently many digits)?

6. (USAMO 1997 modified) Find all positive real numbers x_0 for which there exists an infinite sequence x_0, x_1, x_2, … of real numbers satisfying the recurrence relation

$$x_n = \left\{ \frac{p_n}{x_{n-1}} \right\} \quad \text{for all positive integers } n.$$

(Here $\{x\} = x - \lfloor x \rfloor$ denotes the fractional part of *x*. and p_n denotes the *n*-th prime number.)

7. (IMO 1986) To each vertex of a regular pentagon an integer is assigned in such a way that the sum of all the five numbers is positive. If three consecutive vertices are assigned the numbers *x*, *y*, *z* respectively and $y < 0$ then the following operation is allowed: the numbers *x*, *y*, *z* are replaced by $x + y$, $-y$ and $z + y$ respectively. Such an operation is performed repeatedly as long as at least one of the five numbers is negative. Determine whether this procedure necessarily comes to an end after a finite number of steps.

8. (IMO 2012 shortlist) Several positive integers are written in a row. Iteratively, Alice chooses two adjacent numbers *x* and *y* such that $x > y$ and *x* is to the left of *y*, and replaces the pair (x, y) by either $(y+1, x)$ or $(x-1, x)$. Prove that she can perform only finitely many such iterations.

Topic 5 Combinatorial Proofs

5.1 Binomial identities

There are many identities involving binomial coefficients. In many cases one can work out an *algebraic proof* using either the formula for binomial coefficients or the binomial theorem, but it is often more interesting to work out a *combinatorial proof*. Can you prove the following?

(a) $C_k^n = C_{n-k}^n$

(b) $C_k^n = C_{k-1}^{n-1} + C_k^{n-1}$

(c) $C_0^n + C_1^n + \cdots + C_n^n = 2^n$

(d) $C_1^n + 2C_2^n + \cdots + nC_n^n = n \cdot 2^{n-1}$

(e) $k \cdot C_k^n = n \cdot C_{k-1}^{n-1}$

(f) $C_m^n C_k^m = C_k^n C_{m-k}^{n-k}$

(g) $C_k^{n+m} = C_0^n C_k^m + C_1^n C_{k-1}^m + \cdots + C_k^n C_0^m$

(h) $C_k^k + C_k^{k+1} + \cdots + C_k^n = C_{k+1}^{n+1}$

(i) $(C_0^n)^2 + (C_1^n)^2 + \cdots + (C_n^n)^2 = C_n^{2n}$

(j) $(C_1^n)^2 + 2(C_2^n)^2 + \cdots + n(C_n^n)^2 = nC_{n-1}^{2n-1}$

5.2 Counting in two ways

The combinatorial proofs we have seen in the previous section essentially makes use of *double counting* (or ‘counting in two ways’). By counting the same quantity in two different ways, we get two different expressions, thus establishing their equality.

A simple example is the following. If there is a table of real numbers, then summing each row and adding up the row sums will give the same result as summing each column and then adding up the column sums. This is sometimes known as **Fubini’s theorem**.

Here are two more simple examples illustrating the technique:

- In a party some participants shake hands with each other. Show that the number of people who have shaken hands for an odd number of times is even.
- In a test, each question was correctly answered by more than half of the students. Show that there exists a student who correctly answered more than half of the questions.

5.3 Bijective proofs

Apart from double counting, another common type of combinatorial proof is *bijective proof*. Imagine you are at a party and you would like to determine whether there are more men or women. A simple way to do so will be to ask the participants to form pairs consisting of a man and a woman each. If nobody remains unpaired, then the men and the women have formed a *one-to-one correspondence* (or *bijection*), which would indicate that they have the same number.

Now consider the following two problems.

- How many ways are there to compute the product of n matrices? (For instance there are 2 ways to compute the product ABC, namely, $(AB)C$ and $A(BC)$.)
- How many ways are there to triangulate a convex n-sided polygon?

Work out the answers to the two problems for some small n. Can you give a bijective proof to what you can observe?

5.4 Miscellaneous examples

1. Prove the equality $H_k^n = H_k^{n-1} + H_{k-1}^{n-1} + H_{k-2}^n$.

2. Let $[x]$ denote the greatest integer not exceeding x.

   (a) Find the value of $\left[\log_2 100\right]+\left[\log_3 100\right]+\left[\log_4 100\right]+\cdots+\left[\log_{100} 100\right]$.

   (b) Find the value of $\left[\sqrt{100}\right]+\left[\sqrt[3]{100}\right]+\left[\sqrt[4]{100}\right]+\cdots+\left[\sqrt[100]{100}\right]$.

   (c) Comment on the results of (a) and (b).

3. (IMO 1987) Let $p_n(k)$ be the number of permutations of the set $\{1, 2, \ldots, n\}$ having exactly k fixed points. Prove that

$$\sum_{k=0}^{n} k \cdot p_n(k) = n!$$

4. There are 12 points, no three of which are collinear. Each pair of points is joined by a segment of one of 12 colours. Is it possible to paint the segments in a way such that for any 3 colours, there exists a triangle whose sides are of these 3 colours?

5. (a) There is a set of n men and n women. Let $h(n)$ denote the number of subsets containing the same number of men and women. Find $h(n)$ in terms of n.

   (b) There are also C_n^{2n} ways to choose n people from the set in (a). Can you establish a one-to-one correspondence between a way of choosing a subset of n people and a way of choosing a subset containing the same number of men and women?

   (c) (IMO 2004 Hong Kong Team Selection Test 1) For every positive integer n, let $f(n)$ be the number of all positive integers with exactly $2n$ digits, each having exactly n of the digits equal to 1 and the other n digits equal to 2. Let $g(n)$ be the number of all positive integers with exactly n digits, each of its digits can only be 1, 2, 3 or 4 and the number of 1's equals the numbers of 2's. Prove that $f(n) = g(n)$.

   (d) How would you interpret the fact that $h(n) = f(n) = g(n)$? Can you answer part (b) now?

6. (CHKMO 2004) In a school there are b teachers and c students. Suppose that

- each teacher teaches exactly k students;
- for each pair of distinct students, exactly h teachers teach both of them.

Show that $\frac{b}{h}=\frac{c(c-1)}{k(k-1)}$.

7. A partition of a positive integer n is the expression of n as an unordered sum of one or more positive integers, each of which is called a 'part'. (For example, there are 5 partitions of 4, namely, 4, $3+1$, $2+2$, $2+1+1$ and $1+1+1+1$.) The following lists some special types of partitions of a fixed positive integer n. Identify those with the same size.

(a) Partitions into distinct parts

(b) Partitions into odd parts

(c) Partitions into distinct parts such that each part is at least twice as large as a smaller part

(d) Partitions into distinct odd parts

(e) Partitions in which each part is 1 less than a power of 2 (i.e. 1, 3, 7, 15, …)

(f) Partitions in which the k-th largest part is equal to the number of parts of size at least k

Exercise 5

1. Give combinatorial proofs to the following equalities.

   (a) $C_0^n - C_1^n + C_2^n - \cdots + (-1)^k C_k^n = (-1)^k C_k^{n-1}$

   (b) $C_0^n C_m^n + C_1^n C_{m-1}^{n-1} + \cdots + C_m^n C_0^{n-m} = 2^m C_m^n$

   (c) $\sum_{k=0}^{n} 2^k C_k^n C_{\lfloor \frac{n-k}{2} \rfloor}^{n-k} = C_n^{2n+1}$

   (d) $H_k^n = H_0^n C_k^n + H_1^n C_{k-2}^n + H_2^n C_{k-4}^n + \cdots$

2. (IMC 2002) Two hundred students participated in a mathematical contest. They had 6 problems to solve. It is known that each problem was correctly solved by at least 120 participants. Prove that there must be two participants such that every problem was solved by at least one of these two students.

3. (USAMO 1996) Let a_n be the number of binary sequences of length n containing no three consecutive terms equal to 0, 1, 0 in that order. Let b_n be the number of binary sequences of length n that contain no four consecutive terms equal to 0, 0, 1, 1 or 1, 1, 0, 0 in that order. Prove that $b_{n+1} = 2a_n$ for all positive integers n.

4. There are n points on a circle and some of them are joined by line segments. If no 4-cycle is formed, show that the number of line segments is at most

$$\frac{n}{4}\left(1+\sqrt{4n-3}\right).$$

5. (CHKMO 2007) In a school there are 2007 male and 2007 female students. Each student joins not more than 100 clubs in the school. It is known that any two students of opposite genders have joined at least one common club. Show that there is a club with at least 11 male and 11 female members.

6. (IMO 1998) In a competition, there are a contestants and b judges, where $b \geq 3$ is an odd integer. Each judge rates each contestant as either 'pass' or 'fail'. Suppose k is a number such that, for any two judges, their ratings coincide for at most k contestants. Prove that

$$\frac{k}{a} \geq \frac{b-1}{2b}.$$

7. (IMO 2002) Let n be a positive integer. Let T be the set of points (x, y) in the plane where x and y are non-negative integers with $x + y < n$. Each point of T is coloured red or blue, subject to the following condition: if a point (x, y) is red, then so are all points (x', y') of T with $x' \leq x$ and $y' \leq y$. Let A be the number of ways to choose n blue points with distinct x-coordinates, and let B be the number of ways to choose n blue points with distinct y-coordinates. Prove that $A = B$.

Topic 6 Ramsey Theory

6.1 The classic example

(IMO 1999 HK Selection Test 1) Suppose there are 6 members in an International Mathematical Olympiad team. Prove that among these 6 members there are three members who either all know each other or all don't know each other.

Remarks:

- The bound 6 is sharp. (Why?)
- Because of this, we write $R(3,3)=6$ and call this a **Ramsey number**.
- In general, $R(m,n)$ refers to the smallest positive integer k such that…

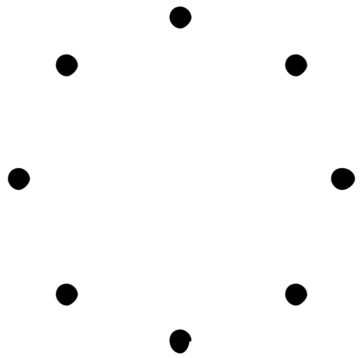
6.2 Values of Ramsey numbers

To establish the value of a Ramsey number, we need to establish both the upper and lower bound. The upper bound is usually established via an *existence argument*, and the lower bound via a *counterexample*. Think about what we did in establishing that $R(3,3)=6$.

Now try to

(a) find the value of $R(2,n)$ for $n \geq 2$;

(b) show that $R(3,4)=9$.



As we can see, finding the exact value of a Ramsey number is very hard in general because the upper and lower bounds have to match. The following is a quote from the famous mathematician Paul Erdös:

> *Suppose an evil alien would tell mankind "Either you tell me [the value of* $R(5,5)$*] or I will exterminate the human race."... It would be best in this case to try to compute it, both by mathematics and with a computer.*
>
> *If he would ask [for the value of* $R(6,6)$*], the best thing would be to destroy him before he destroys us, because we couldn't [determine* $R(6,6)$*].*

Indeed, not many Ramsey numbers are known. Effort is mainly made on establishing bounds on Ramsey numbers. The following table shows some known Ramsey numbers (in bold) and the most updated bounds (as at October 2020) on some Ramsey numbers.

m \ n	3	4	5	6	7	8	9	10	11	12
3	**6**	**9**	**14**	**18**	**23**	**28**	**36**	40 42	46 50	52 59
4		**18**	**25**	36 41	49 61	59 84	73 115	92 149	97 191	128 238
5			43 48	58 87	80 143	101 216	133 316	149 442	158 633	184 871
6				102 165	115 298	134 495	183 780	204 1171	262 1804	263 2675

6.3 Generalising Ramsey numbers

There are basically two directions to generalise Ramsey numbers:

- Increase the number of colours
- Rewrite $R(m,n)$ as $R(K_m, K_n)$ and generalise it to $R(G,H)$

The following two examples illustrate the above generalisations:

- (IMO 1964) Seventeen people correspond by mail with one another – each one with all the rest. In their letters only three different topics are discussed. Each pair of correspondents deals with only one of these topics. Prove that there are at least three people who write to each other about the same topic.

- (CGMO 2005 HK Selection Quiz 1) N people attend a meeting, and some of them shake hands with each other. Suppose that

  (i) each person shakes hands with at most 100 other people, and

  (ii) among any 50 people, there exist at least two who have shaken hands with each other.

  Find the greatest possible value of N.

6.4 Miscellaneous examples

1. (CWMO 2005) There are n new students. Among any three of them there exist two who know each other, and among any four of them there exist two who do not know each other. Find the greatest possible value of n.

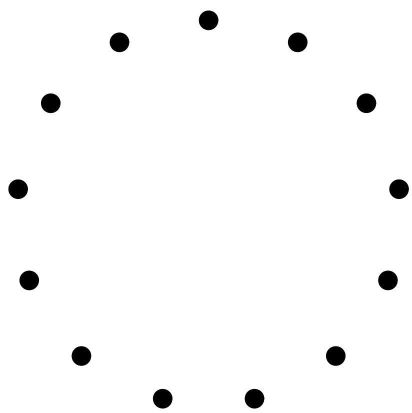
2. Let m and n be integers greater than 2.

   (a) Show that $R(m,n) \le R(m-1,n) + R(m,n-1)$.

   (b) Using (a), deduce that $R(m,n) \le C_{m-1}^{m+n-2}$.

   (c) When we can replace the '≤' sign in (a) by the '<' sign?

3. Prove that $R(3,5) = 14$.



4. (CHKMO 2005) On a planet there are $3\times 2005!$ aliens and 2005 languages. Each pair of aliens communicates with each other in exactly one language. Show that there are 3 aliens who communicate with each other in one common language.

5. Each of the integers 1, 2, 3, …, 325 is coloured red or blue. Show that there exist three numbers of the same colour which form an arithmetic sequence.

6. Let $k \geq 3$ be a fixed positive integer.

(a) If n is a positive integer such that $C_k^n \cdot 2^{1-C_2^k} < 1$, prove that $R(k,k) > n$.

(b) Hence or otherwise, deduce that $R(k,k) > 2^{\frac{k}{2}}$.

Exercise 6

1. Prove that $R(4,4)=18$.

2. Prove that for $k \geq 3$, $2k \leq R(3,k) \leq \frac{k^2+3}{2}$.

3. (APMO 2003) Given two positive integers *m* and *n*, find the smallest positive integer *k* such that among any *k* people, either there are 2*m* of them who form *m* pairs of mutually acquainted people or there are 2*n* of them forming *n* pairs of mutually unacquainted people.

4. Let $n = 7\left(2\cdot 3^7+1\right)\left[2\cdot 3^{7(2\cdot 3^7+1)}+1\right]$.

   If each of the integers 1, 2, 3, …, *n* is coloured red, yellow or blue, show that there exist three numbers of the same colour which form an arithmetic sequence.

5. Let $W(k)$ denote the smallest positive integer *n* such that if each of the integers 1, 2, …, *n* is coloured red or blue, there are *k* numbers of the same colour which form an arithmetic sequence . Prove that $W(k) > 2^{\frac{k}{2}}$.

Topic 7   Graph Theory

7.1   Basics

Many problems in combinatorics can be phrased using the language of *graph theory*. A **graph** is usually drawn with some *points* (formally, **vertices**) and some *line segments* (formally, **edges**) joining some pair of points.

- Unless otherwise specified, all graphs are assumed to be finite (i.e. there are finitely many vertices) and **simple** (i.e. there are no *loops* and no *multiple edges*). We sometimes write $G = (V, E)$ where G denotes a graph, V denotes its set of vertices and E denotes its set of edges.
- The **degree** of a vertex is the number of edges adjacent to it.
- The **girth** of a graph is the length of the smallest *cycle* (and defined to be infinity if no cycle exists).
- A graph is **connected** if each pair of points is connected by *path*. If a graph is not connected, then it is divided into two or more **connected components** (or simply **components**).
- The **distance** between two points in a connected graph is the length of the shortest path joining them. The **diameter** of a connected graph is the maximum of the distances between two points.

Here is some quick exercise:

- Show that there is an even number of vertices in a graph with odd degree.
- If a graph has minimum degree $k \geq 2$, show that it contains a path of length at least k and a cycle of length at least $k+1$.
- Let G be a graph with n vertices and m edges. If $m \geq n$, show that G contains a cycle.
- Show that a k-regular graph of girth 5 and diameter 2 has exactly k^2+1 vertices.
  (Remark: It has been shown that if such a graph exists, k can only be 2, 3, 7 or possibly 57.)

7.2 Eulerian and Hamiltonian graphs

Two important classes of graphs are Eulerian graphs and Hamiltonian graphs.

- A graph is said to be **Eulerian** if it contains an *Euerlian circuit*, which is a *circuit* containing all edges of the graph.
- A graph is said to be **Hamiltonian** if it contains a *Hamilton cycle*, which is a cycle passing through all vertices of the graph.

The Eulerian problem is essentially the same as the classic 'single pen-stroke problem'. It is easy to show that a graph is Eulerian if and only if every vertex has even degree.

The Hamiltonian problem, on the other hand, is much harder. There is no simple equivalent condition for Hamiltonicity. There are however some known conditions there are either necessary or sufficient.

- (Necessary condition) Let G be a Hamiltonian graph with n vertices. Show that on deleting k vertices and their associated edges, where $1 \le k < n$, the resulting graph has at most k connected components.
- (Sufficient condition) Let G be a graph on n vertices. If each vertex has degree at least $\frac{n}{2}$, show that G is Hamiltonian.

As an exercise, try the following:

For positive integers m, n, let $G_{m,n}$ be the bipartite graph with bipartition (X,Y) where $X=\{a_1,a_2,\dots,a_m,b_1,b_2,\dots,b_n\}$ and $Y=\{c_1,c_2,\dots,c_m,d_1,d_2,\dots,d_n\}$ such that

- a_i is adjacent to c_i for each $i=1,2,\dots,m$ (so for instance a_1 is adjacent to c_1 but not c_2)
- a_i is adjacent to d_j for all i, j
- b_i is adjacent to c_j for all i, j

Determine all pairs of (m,n) for which $G_{m,n}$ is Hamiltonian.

7.3 Matching in bipartite graphs

- A graph is **bipartite** if the vertex set can be partitioned into two subsets such that every edge has endpoints in different subsets.
- A **matching** in a graph is a set of edges, no two of which share a common endpoint. The endpoints of the edges in a matching are said to be **saturated**. A **perfect matching** is a matching which saturates every vertex.
- A k-regular graph is said to be **1-factorable** if its edge set can be partitioned into k perfect matchings.

One of the most famous results concerning matching is **Hall's theorem**, which states:

A bipartite graph G with bipartition (X, Y) contains a matching which saturates every vertex in X if and only if $\|N(S)\| \geq \|S\|$ for every $S \subseteq X$, where $N(S)$ denotes the set of vertices which are adjacent to at least one vertex in S.

Some matching problems:

1. Let G be a k-regular bipartite graph where $k > 0$. Show that

   (a) G has a perfect matching;

   (b) G is 1-factorable.

2. (Canada MO 2006, slightly rephrased) In some cells of an $m \times n$ table a positive real number is written. There is at least one number in each row and in each column. Whenever there is such a number x in a cell, the sum of the numbers in the row containing x must be equal to the sum of the numbers in the column containing x. Show that $m = n$.

7.4 Miscellaneous examples

1. A finite decreasing sequence of non-negative integers is said to be graphic if it is the degree sequence of a (simple) graph. Determine whether each of the following sequences is graphic.

   (a) (6,5,4,4,3,3,2)

   (b) (6,6,5,4,3,3,2)

   (c) (6,6,5,4,3,3,1)

2. Some obvious necessary conditions for a decreasing sequence $(d_1, d_2, \ldots, d_n)$ to be graphic are: each term is between 0 and $n-1$, the sum of all terms is even, etc. Show that each of the following is also a necessary condition.

   (a) At least two terms are the same.

   (b) For each $1 \le k \le n$ we have
$$d_1 + d_2 + \cdots + d_k \le k(k-1) + \min\{k, d_{k+1}\} + \cdots + \min\{k, d_n\}.$$

3. If a graph on n vertices is C_4-free, show that it has at most $\frac{n}{4}\left(1+\sqrt{4n-3}\right)$ edges.

4. On the circumference of a circle 100 points are marked. Now each pair of points is joined by a line segment. What is the maximum number of regions into which these segments can divide the circle?

5. Let G be a graph in which any two odd cycles share at least one common vertex. Show that G is 5-colourable.

Exercise 7

1. (JMO 1998) In a country, there are 1998 airports. For any three airports *A*, *B*, *C*, at least one of the pairs *AB*, *BC*, *CA* is not connected by an airline. (The airlines operate direct flights only.) How many airlines are there at most?

2. (APMO 1989) Let *S* be a set consisting of *m* pairs (*a*, *b*) of positive integers with the property that $1 \le a < b \le n$. Show that there are at least $\frac{4m}{3n}\left(m - \frac{n^2}{4}\right)$ triples (*a*, *b*, *c*) such that (*a*, *b*), (*a*, *c*) and (*b*, *c*) belong to *S*.

3. (IMO 1991) Suppose *G* is a connected graph with *k* edges. Prove that it is possible to label the edges 1, 2, 3, …, *k* in such a way that at each vertex which belongs to two to more edges then greatest common divisor of the integers labelling those edges is equal to 1.

4. (IMO 1992) Consider nine points in space, no four of which are coplanar. Each pair of points is joined by an edge (that is, a line segment) and each edge is either coloured blue or red or left uncoloured. Find the smallest value of *n* such that whenever exactly *n* edges are coloured, the set of coloured edges necessarily contains a triangle all of whose edges have the same colour.

5. (IMO 2019) A social network has 2019 users, some pairs of whom are friends. Whenever user *A* is friends with user *B*, user *B* is also friends with user *A*. Events of the following kind may happen repeatedly, one at a time:

   Three users *A*, *B*, and *C* such that *A* is friends with both *B* and *C*, but *B* and *C* are not friends, change their friendship statuses such that *B* and *C* are now friends, but *A* is no longer friends with *B*, and no longer friends with *C*. All other friendship statuses are unchanged.

   Initially, 1010 users have 1009 friends each, and 1009 users have 1010 friends each. Prove that there exists a sequence of such events after which each user is friends with at most one other user.

6. (APMO 1990) A set of 1990 persons is divided into non-intersecting subsets in such a way that

- no one in a subset knows all the others in the subset;
- among any three persons in a subset, there are always at least two who do not know each other;
- for any two persons in a subset who do not know each other, there is exactly one person in the same subset knowing both of them.

(a) Prove that within each subset, every person has the same number of acquaintances.

(b) Determine the maximum possible number of subsets.

7. (IMO 2007) In a mathematical competition some competitors are friends; friendship is always mutual. Call a group of competitors a *clique* if each two of them are friends. The number of members in a clique is called its *size*.

It is known that the largest size of cliques is even. Prove that the competitors can be arranged in two rooms such that the largest size of cliques in one room is the same as the largest size of cliques in the other room.

8. (SMMC 2019) Let G be a finite simple graph and let k be the largest number of vertices of any clique in G. Suppose that we label each vertex of G with a non-negative real number, so that the sum of all such labels is 1. Define the value of an edge to be the product of the labels of the two vertices at its ends. Define the value of a labelling to be the sum of the values of the edges.

Prove that the maximum possible value of a labelling of G is $\frac{k-1}{2k}$.